

Notes on holomorphic line bundles over a torus I

- References :
- John Baez, Line Bundles on Complex Tori (Part 1)
 - Birkenhake and Lange, Complex Abelian Varieties
 - Brylinski, Loop spaces, characteristic classes and geometric quantization (chapter 2)
 - Le Floch Chapter 3
 - Polishchuk, Abelian varieties, theta functions and the Fourier transform.

NB

But most relevant reference: Schottenlocher, Lecture notes on geometric quantization, especially Appendix E.

Firstly, smooth line bundles over a manifold M are classified by their transition functions. A more fancy way to say this is:

Theorem 1 There is a canonical isomorphism of abelian groups

$$\left\{ \text{smooth complex line bundles over } M \right\} / \text{isomorphism} \cong \check{H}^1(M, \underline{\mathbb{C}}^*)$$

Čech cohomology
(reminder below)

This is an abelian group under $\otimes$.

Inverses are $[L^{-1}] = [L^\vee]$ since $L^\vee \otimes L \cong \mathbb{C}$

↑
dual line bundle.

the sheaf on $\mathcal{O}_P(M)^{op}$
which sends
 $U \mapsto C^\infty(U, \mathbb{C}^*)$.

Recall the definition of Čech cohomology associated to an open cover $(U_i)_{i \in I}$ of M :

$$\check{H}^n(X, \mathcal{U}, F)$$

$\uparrow$
 open cover
of X

$\nwarrow$ a presheaf of abelian groups
on X , i.e. a functor
 $F: \text{Op}(X)^{\text{op}} \longrightarrow \text{Ab}$
 $\uparrow$
 poset of open sets on X
 ordered by inclusion

It is the cohomology of the cochain complex associated to the simplicial set given by the nerve N of the open cover $\mathcal{U} = (U_i)_{i \in I}$:

$$N_n = \left\{ (i_0, \dots, i_n) \in I^n \text{ such that } U_{i_0} \cap \dots \cap U_{i_n} \neq \emptyset \right\}$$

The boundary maps are

$$\partial_k : N_n \longrightarrow N_{n-1} \quad (k = 0 \dots n)$$

$$(i_0, \dots, i_n) \longmapsto (i_0, \dots, \hat{i}_k, \dots, i_n).$$

And the degeneracy maps are:

$$s_j : N_{n-1} \longrightarrow N_n \quad (j = 0 \dots n-1)$$

$$(i_0, \dots, i_{n-1}) \longmapsto (i_0, \dots, i_j, i_j, \dots, i_{n-1}).$$

So, the cochain complex is

$$C^n(M, \mathcal{U}, F) := \bigoplus_{(i_0, \dots, i_n) \in N_n} F(U_{i_0} \cap \dots \cap U_{i_n})$$

with coboundary maps

$$d: C^n(M, \mathcal{U}, F) \longrightarrow C^{n+1}(M, \mathcal{U}, F)$$

given by $\sum_{k=0}^n (-1)^k F(\partial_k)$ as usual.

E.g.

- $C^0 = \bigoplus_{i: U_i \neq \emptyset} F(U_i)$

Let $a = \{a_i \in F(U_i)\}$

$(da)_{i,j} = (a_i - a_j) \Big|_{U_i \cap U_j} \in F(U_i \cap U_j)$
- $C^1 = \bigoplus_{(i,j): U_i \cap U_j \neq \emptyset} F(U_i \cap U_j)$

Let $a = \{a_{i,j} \in F(U_i \cap U_j)\}$

$(da)_{i,j,k} = (a_{j,k} - a_{i,k} + a_{i,j}) \Big|_{U_i \cap U_j \cap U_k} \in F(U_i \cap U_j \cap U_k)$
- $C^2 = \bigoplus_{(i,j,k): U_i \cap U_j \cap U_k \neq \emptyset} F(U_i \cap U_j \cap U_k)$

So for instance, a 0-cocycle $a = (a_i)_{i \in I}$ is closed if

$$da = 0 \iff a_i \Big|_{U_i \cap U_j} = a_j \Big|_{U_i \cap U_j} \text{ for all } U_i, U_j \text{ with } U_i \cap U_j \neq \emptyset.$$

And so on. So we've now defined

another notation $\longrightarrow$
for Čech cohomology $\check{H}^1(M, \mathcal{U}, \mathbb{C}^*)$

The "actual" cohomology is the colimit over all open covers $\mathcal{U}$:

$$H_{\check{\text{Cech}}}^n(M, \mathcal{F}) := \operatorname{colim}_{\mathcal{U}} H_{\check{\text{Cech}}}^n(M, \mathcal{F}, \mathcal{U})$$

- but remember:
- (1) by definition, for any cover $\mathcal{U}$, an element of $H^n(M, \mathcal{F}, \mathcal{U})$ induces an element of the colimit space
 - (2) If $\mathcal{U}$ is a good cover (i.e. if all intersections $U_{i_0} \cap \dots \cap U_{i_n}$ are contractible) then this map

$$H^n(M, \mathcal{F}, \mathcal{U}) \longrightarrow H^n(M, \mathcal{F})$$

is an isomorphism.

Finally, in all the cases we deal with, Čech cohomology of $(M, \mathcal{F})$

(computed according to an open cover, as above) is isomorphic to sheaf

cohomology of $(M, \mathcal{F})$ (which uses projective resolutions). Basically, you can think of the nerve of a good cover as providing a nice projective resolution, in which you can compute sheaf cohomology. So I will often use "Čech cohomology" and "Sheaf cohomology" interchangeably.

Ok. So let's unravel

$$H^1(M, \underline{\mathbb{C}}_M^*)$$

We fix a good open cover $\mathcal{U} = (U_i)_{i \in I}$.

A 2-cocycle is a collection $g = (g_{ij})$ of smooth functions

$$g_{ij} : U_i \cap U_j \longrightarrow \mathbb{C}^*, \quad U_i \cap U_j \neq \emptyset$$

such that:

$$g_{ij} g_{jk} g_{ki} = 1.$$

(We can also ask that $g_{ii} = 1$ and $g_{ij} g_{ji} = 1$).

We have $g \sim g'$ if there exists $h = (h_i : U_i \longrightarrow \mathbb{C}^*)_{i \in I}$ that

$$g'_{ij} = g_{ij} \cdot \frac{h_i}{h_j}.$$

The theorem we had before is now clear:

Theorem There is a canonical isomorphism of abelian groups

$$\left\{ \begin{array}{c} \text{Complex line bundles over} \\ M \end{array} \right\} / \text{isomorphism} \cong \check{H}^1(M, \underline{\mathbb{C}}_M^*)$$

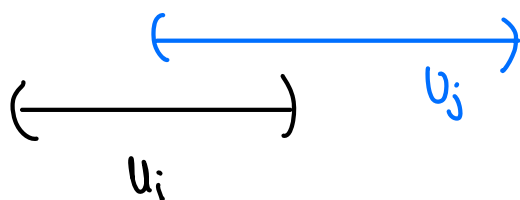
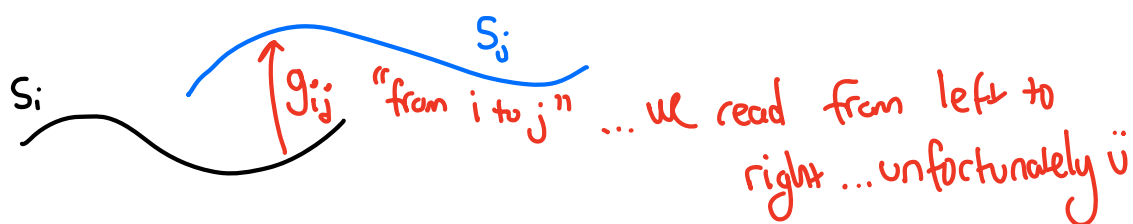
Proof Given a line bundle $L \rightarrow M$, we choose nowhere-vanishing smooth sections

$$s_i : U_i \longrightarrow L|_{U_i}$$

from which we get a collection of transition functions g_{ij} on U_{ij} via

$$s_j(x) = \underbrace{g_{ij}(x)}_{\in \mathbb{C}^*} s_i(x) \quad \text{on } U_{ij}$$

$$s_j = g_{ij} s_i$$



If $\phi : L \xrightarrow{\cong} L'$ is an isomorphism, then we can write

$$\phi(s_i(x)) = \underbrace{h_i(x)}_{\in \mathbb{C}^*} s'_i(x)$$

And we will have

$$g'_{ij} = g_{ij} \frac{h_i}{h_j} \quad \text{on } U_i \cap U_j.$$

In the reverse direction, given transition functions (g_{ij}) , we build a line bundle L as the quotient

$$L := \frac{\bigsqcup_{i \in I} U_i \times \mathbb{C}}{\sim}$$

where $(i, x, z) \sim (j, x, g_{ij}(x)z)$.

□

(For a fuller, more pedestrian treatment, see Le Floch 3.1)
 " high-brow " Brylinski 2.1

We

$$\text{If } S_j = g_{ij} S_i$$

$$S_j = g_{kj} S_k \quad (\text{Schwartz})$$

Then a section will be

$$f_i \in U_i \quad \text{s.t.}$$

$$\underbrace{\hspace{2cm}}_{S_i} S_j$$

$$\underbrace{\hspace{2cm}}_{S_i} f_i S_i = f_j S_j$$

$$\text{i.e. } f_i S_i = f_j g_{ij} S_i$$

$$f_i = f_j \cdot g_{ij}$$

Ok. That's what a smooth line bundle is, and their classification. Similarly we have, by exactly the same proof:

Theorem If M is a complex manifold, then There is a canonical isomorphism of abelian groups

$$\left\{ \text{holomorphic complex line bundles over } M \right\} / \text{isomorphism} \cong \check{H}^1(M, \mathcal{O}_M^*)$$

the sheaf which
sends $U \mapsto$ nonvanishing
holomorphic
functions on U .

Example Compute the transition functions for, and holomorphic sections of, the tautological bundle τ over $\mathbb{CP}^1$.

Recall $\tau_\ell = \ell$ where ℓ is a line through origin in $\mathbb{C}^2$.

Write $\mathbb{C} \cdot (z_0, z_1)$ for the complex line through (z_0, z_1) in $\mathbb{C}^2$.

So

$$\tau_{[z_0:z_1]} = \mathbb{C} \cdot (z_0, z_1)$$

Transition functions : Cover $\mathbb{CP}^1$ by.

$$U_0 = \{ [z_0 : z_1] : z_0 \neq 0 \}$$

$$U_1 = \{ [z_0 : z_1] : z_1 \neq 0 \}$$

We get nowhere vanishing holomorphic sections

$$s_0 : U_0 \longrightarrow \tau|_{U_0}$$

$$[z_0 : z_1] \longmapsto \left(1, \frac{z_1}{z_0} \right) \in \mathbb{C} \cdot (z_0, z_1)$$

$$s_1 : U_1 \longrightarrow \tau|_{U_1}$$

$$[z_0 : z_1] \longmapsto \left(\frac{z_0}{z_1}, 1 \right) \in \mathbb{C} \cdot (z_0, z_1)$$

Over $U_0 \cap U_1$, we have:

$$\underbrace{s_1([z_0 : z_1])}_{= \left(\frac{z_0}{z_1}, 1 \right)} = \frac{z_0}{z_1} \underbrace{s_0([z_0 : z_1])}_{= \left(1, \frac{z_1}{z_0} \right)}$$

s_0

$$g_{01}([z_0 : z_1]) = \frac{z_0}{z_1}$$

on $U_0 \cap U_1$.

Ok. What is a global holomorphic section $s \in H^0(\mathbb{CP}^1, \mathcal{K})$?

Well, over U_0 and U_1 we can write

$$s|_{U_0} = f_0 \cdot s_0 \quad \text{on } U_0$$

$$s|_{U_1} = f_1 \cdot s_1 \quad \text{on } U_1$$

defining holomorphic functions $f_i : U_i \longrightarrow \mathbb{C}$. In order for this to be consistent over $U_0 \cap U_1$, we need

$$f_0 s_0 = f_1 s_1 \quad \text{on } U_0 \cap U_1$$

i.e.

$$f_0 s_0 = f_1 g_{01} s_0$$

i.e.

$$\boxed{f_1 = g_{01}^{-1} f_0 \quad \text{on } U_0 \cap U_1} \quad (*)$$

Ok. Now remember the holomorphic charts for $\mathbb{P}^1$:

$$\phi_0 : U_0 \xrightarrow{\cong} \mathbb{C}$$

$$[z_0 : z_1] \longmapsto z := \frac{z_1}{z_0}$$

$$[1 : z] \longleftarrow z$$

$$\phi_1 : U_1 \xrightarrow{\cong} \mathbb{C}$$

$$[z_0 : z_1] \longmapsto w := \frac{z_0}{z_1}$$

$$[w : 1] \longleftarrow w$$

On $U_0 \cap U_1$, these coordinates are related via

$$w = \frac{1}{z}.$$

Writing $F_i := f_i \circ \phi_i^{-1}$ for the coordinate descriptions of f_0 and f_1 , we see that a section of τ amounts to a pair of holomorphic functions

$$F_0 : \mathbb{C} \longrightarrow \mathbb{C} \\ z \longmapsto F_0(z)$$

$$F_1 : \mathbb{C} \longrightarrow \mathbb{C} \\ w \longmapsto F_1(w)$$

(which, since they are entire holomorphic functions on $\mathbb{C}$, we can expand as

$$F_0(z) = a_0 + a_1 z + a_2 z^2 + \dots$$

$$F_1(z) = b_0 + b_1 z + b_2 z^2 + \dots,$$

)

such that $(*)$ holds, i.e. writing $[1:z]$ ($z \neq 0$) for an arbitrary element of $U_0 \cap U_1$, we have

$$\underbrace{f_1([1:z])}_{\substack{= f_1([\frac{1}{2}:1]) \\ = F_1(\frac{1}{2})}} = \underbrace{g_{01}^{-1}([1:z])}_{= z} \underbrace{f_0([1:z])}_{= F_0(z)} \quad \text{on } \mathbb{C} \setminus \{0\}$$

So we need

$$b_0 + \frac{b_1}{z} + \frac{b_2}{z^2} + \dots = z \left(a_0 + a_1 z + a_2 z^2 + \dots \right)$$

$$= a_0 z + a_1 z^2 + a_2 z^3 + \dots$$

These are both to be thought of as Laurent series on $\mathbb{C} \setminus \{0\}$.

They are equal iff their corresponding coefficients are equal.

$$\Rightarrow a_n = b_n = 0 \text{ for all } n.$$

So,

$$H^n(\mathbb{CP}^1, \tau) = \{0\}.$$

How about $\tau^\vee$? Well, the transition functions for $L^\vee$ are the inverse of the transition functions for L (exercise). Hence, a holomorphic section of $\tau^\vee$ corresponds to a pair of entire holomorphic functions

$$F_0 : \mathbb{C} \longrightarrow \mathbb{C}$$

$$z \longmapsto a_0 + a_1 z + \dots$$

$$F_1 : \mathbb{C} \longrightarrow \mathbb{C}$$

$$z \longmapsto b_0 + b_1 z + \dots$$

Such that on $\mathbb{C} \setminus \{0\}$ we have

$$F_1\left(\frac{1}{z}\right) = \frac{1}{z} F_0(z)$$

i.e. $b_0 + \frac{b_1}{z} + \frac{b_2}{z^2} + \dots = \frac{1}{z} \left(a_0 + \frac{a_1}{z} + \frac{a_2}{z^2} + \dots \right)$ on $\mathbb{C} \setminus \{0\}$

$\therefore b_0 = a_1, a_1 = b_0$ and all other a_n, b_n vanish.

So $H^0(\mathbb{CP}^1, \mathcal{O}^v)$ is 2-dim.

□

Ok. Now, I want to exhibit an isomorphism

$$\check{H}^1(M, \underline{\mathbb{C}}_M^*) \cong \check{H}^2(M, \underline{\mathbb{Z}})$$

↑ the sheaf which assigns
 $U \mapsto \left\{ \begin{array}{l} \text{locally constant functions} \\ f: U \rightarrow \mathbb{Z} \end{array} \right\}$

The highbrow way to do this is to appeal to the fact that we have a short exact sequence of sheaves: (see Brylinski 1.1.9)

$$0 \longrightarrow \underline{\mathbb{Z}} \hookrightarrow \underline{\mathbb{C}}_M \xrightarrow{\exp(2\pi i \cdot)} \underline{\mathbb{C}}_M^* \longrightarrow 0 \quad (*)$$

That is to say, for every $x \in M$ we have an exact sequence of abelian groups given by the stalks of the sheaves at x :

$$0 \longrightarrow \underline{\mathbb{Z}}_x \longrightarrow \underline{\mathbb{C}}_{M,x} \xrightarrow{\exp(2\pi i \cdot)} \underline{\mathbb{C}}_{M,x}^* \longrightarrow 0$$

Remember, the stalk $\mathcal{F}_x$ of a sheaf $\mathcal{F}$ at $x \in M$ is

the set of germs of sections of F at x , i.e.

$$F_x = \frac{\bigsqcup_{\text{open neighborhoods } U \text{ of } x} F(U)}{\sim}$$

where $(U, s \in F(U)) \sim (U', s' \in F(U'))$ if there exists an open set $V \subseteq U \cap U'$, containing x , such that

$$s|_V = s'|_V.$$

So: Some equation between sheaves E and F holds "at the level of the stalks" when it holds on small enough open sets U .

Now, The only nontrivial part of $(*)$ is the surjectivity of

$$\mathbb{C}_{M \times} \xrightarrow{\exp(2\pi i \cdot)} \mathbb{C}_{M \times}^*$$

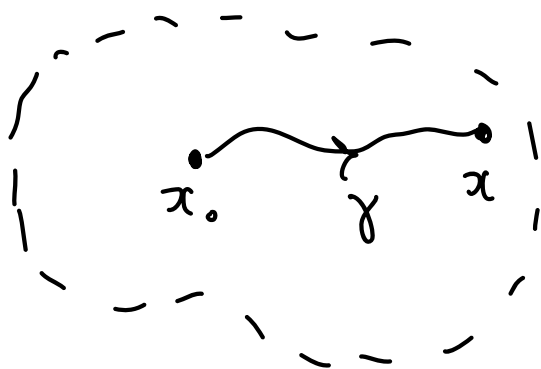
What this is saying is that, on a small enough open neighborhood U of x , the map

$$\left\{ \begin{array}{l} \text{holomorphic functions} \\ \text{on } U \end{array} \right\} \xrightarrow{\exp(2\pi i \cdot)} \left\{ \begin{array}{l} \text{nowhere vanishing} \\ \text{holomorphic functions on } U \end{array} \right\}$$

$$f \longmapsto e^{2\pi i f}$$

is surjective. This is true, because we can take U to be contractible, so we can get a logarithm of $g \in \text{Hol}(U, \mathbb{C}^*)$ on the whole of U . Concretely, fix $x_0 \in U$, and then define $f(x)$ by integrating the logarithmic derivative of g along any path γ from x_0 to x (different paths are homotopic, hence give same answer since $\oint df = 0$)

$$f(x) = \frac{1}{2\pi i} \int_{x_0 \xrightarrow{\gamma} x} \frac{g'(z)}{g(z)} dz$$



$$\left(\text{Remember : } \int \frac{1}{z} dz = \log z \right)$$

Ok. Now, the magic of homological algebra is that if we have a short exact sequence of "coefficient groups" (in our case, sheaves)

$$0 \longrightarrow \underline{\mathbb{Z}} \hookrightarrow \underline{\mathbb{C}}_M \xrightarrow{\exp(2\pi i \cdot)} \underline{\mathbb{C}}_M^* \longrightarrow 0 \quad (*)$$

then we get a long exact sequence in cohomology for any M :

$$\begin{aligned} H^0(M, \underline{\mathbb{Z}}) &\longrightarrow H^0(M, \underline{\mathbb{C}}_M) \longrightarrow H^0(M, \underline{\mathbb{C}}_M^*) \\ &\quad \delta \\ &\hookrightarrow H^1(M, \underline{\mathbb{Z}}) \longrightarrow H^1(M, \underline{\mathbb{C}}_M) \longrightarrow H^1(M, \underline{\mathbb{C}}_M^*) \\ &\quad \delta \\ &\hookrightarrow H^2(M, \underline{\mathbb{Z}}) \longrightarrow H^2(M, \underline{\mathbb{C}}_M) \longrightarrow H^2(M, \underline{\mathbb{C}}_M^*) \\ &\quad \delta \\ &\hookrightarrow \dots \end{aligned}$$

The magic is those mysterious connecting maps δ , which you learnt how to construct by diagram chasing in homological algebra (we'll work it out concretely in our setting, in a moment). Let's focus on the following segment of the long exact sequence:

$$\underline{H^1(M, \underline{\mathbb{C}}_M)} \longrightarrow H^1(M, \underline{\mathbb{C}}_M^*) \xrightarrow{\delta} H^2(M, \underline{\mathbb{Z}}) \longrightarrow \underline{H^2(M, \underline{\mathbb{C}}_M)}$$

=0

=0

The blue groups $H^i(M, \underline{\mathbb{C}}_M)$ are zero because the sheaf $\underline{\mathbb{C}}_M$ is "soft" (see Brylinski 1.4) ... this amounts to saying that smooth manifolds admit partitions of unity. Therefore, the connecting map is an isomorphism:

$$\delta: H^1(M, \underline{\mathbb{C}}_M^*) \xrightarrow{\cong} H^2(M, \underline{\mathbb{Z}})$$

Let's unravel the details here. (Somewhere in Bott and Tu's classic book on "Differential forms in algebraic topology", Bott says that the mark of a true topologist is being able to explicitly unwrap what these connecting homomorphisms are doing).

We start with a cocycle $g \in Z^1(M, \underline{\mathbb{C}}_M^*)$, i.e.

$$g_{ij}: U_i \cap U_j \longrightarrow \mathbb{C}^*.$$

We always work on a good open cover, so $U_i \cap U_j$ is contractible, so we can find logs of g_{ij} , i.e. we can write

$$g_{ij}(x) = e^{2\pi i f_{ij}(x)}$$

for some smooth functions $f_{ij} : U_{ij} \rightarrow \mathbb{C}$. Now,

$$\left. \begin{aligned} g_{ij} g_{jk} g_{ki} &= 1 \\ e^{2\pi i (f_{ij} + f_{jk} + f_{ki})} &= 1 \end{aligned} \right\} \text{ on } U_i \cap U_j \cap U_k$$

i.e.

We can't say that the exponent is zero, only that it is an integer, i.e.

$$\underbrace{f_{ij}(x) + f_{jk}(x) + f_{ki}(x)}_{=: n_{ijk}} \text{ is locally constant on } U_i \cap U_j \cap U_k$$

Now, I claim that these (n_{ijk}) are a cocycle in $Z^2(M, \mathbb{Z})$.

Why? Well, I need to show that

$$n_{jke} - n_{ike} + n_{ije} - n_{ijk} = 0 \quad \text{on } U_i \cap U_j \cap U_k \cap U_e$$

I know:

$$df_{ij}(x) + df_{jk}(x) + df_{ki}(x) = 0 \quad \text{on } U_i \cap U_j \cap U_k$$

... this is some magic here. More next time!

□